

Case Study

A strategic website evaluation of online travel agencies

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ARTICLE INFO

Article history:

Received 23 May 2010

Accepted 15 December 2010

Keywords:

Website evaluation

Online travel agencies

Strategy consistency

ABSTRACT

Online travel Web sites have been the most frequently visited online information facilities by travelers. To evaluate the effectiveness of a travel Web site, the Web site manager should regularly check whether or not it is fulfilling the objectives that were established for it. This research uses a strategic Web site evaluation framework to introduce a five-stage process for examining the consistency of Web site's presence and its intended strategies. Two leading online travel agencies with different business strategies are selected to demonstrate methods of implementing a strategic evaluation framework and to compare the evaluation results. A hierarchical evaluation structure is introduced to explicitly delineate the two Web sites' different strategy intentions and related evaluation criteria. Results show that an individual Web site's strategy-inconsistent criteria can be easily identified through a gap analysis and criteria performance matrix. A strategy-inconsistent dimension can be discovered through a radar chart analysis of the 4PsC (Product, Promotion, Price, Place, and Customer Relationship) dimensions and a transaction phases analysis.

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1. Introduction

Along with the rapidly increasing popularity of the Internet, travel Web sites have become some of the most frequently visited online information facilities by travel planners (Choi, Lehto, & O'leary, 2007; Law & Leung, 2000; Zhou & DeSantis, 2005). The Internet is leading businesses into a new era in the field of communication and is changing business transactions. The market of products and services in the tourism industry relies heavily on information and has a highly segmented structure (Roney & Özturan, 2006; Thorn & Chen, 2005). In fact, the tourism industry is one of the world's largest industries adopting the Internet as the medium for e-business revolution. Moreover, the Web is now the most widely used tool in conducting research on tourist information and promoting regional tourism; it is also cheaper compared with other forms of promotion and advertising (Horng & Tsai, 2010; Standing & Vasudavan, 2000; Stepchenkova, Tang, Jang, Kirilenko, & Morrison, 2010). Maintaining an effective Web site is thus vital for a business to strengthen its customer relationships and enlarge its market segment (Law, Qi, & Buhalis, 2010).

There are only a few successful studies on what motivates users to browse and make purchases on travel Web sites (Law et al., 2010). Several review studies have noted that there is no universally accepted technique or standard for Web site evaluation (Law et al., 2010; Morrison, Taylor, & Douglas, 2004; Tsai, Chou, & Lai, 2010). Although the findings of these studies have revealed the most crucial and prevalent features of successful tourism Web sites in general, these identified features may not necessarily be applicable to every Web site because organizations have their own Web site development strategies to attain their goals and objectives. Web sites are developed based on organizational goals and objectives. Therefore, they should be reviewed regularly to determine whether they are fulfilling the reasons for which they were developed, and it is not necessary to have a universal standard in assessing the success or effectiveness of a Web site because each Web site is designed for a specific reason (Clyde, 2000).

Many previous studies have adopted user surveys in investigating user perceptions of selected Web sites, whereas only a few studies have adopted expert-based evaluations. User-based surveys can be regarded as an external evaluation method to examine whether a Web site is "doing the thing right" in meeting user expectations. Nevertheless, to examine whether a Web site is "doing the right thing" in meeting its Web strategy requirements, an internal evaluation should be conducted by a panel of experts before the conduct of external surveys. The internal evaluation should persist until the Web site becomes consistent with its

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strategy, after which an external evaluation may follow. Thus, this study aims to use a strategic Web site evaluation framework to develop a systemic evaluation process in examining the consistency of Web site presence and intended strategy. The following section reviews recent studies on Web site evaluation. Section 3 introduces our proposed strategic evaluation methodology. In Section 4, the methodology is applied to two leading travel Web sites in Taiwan. The last section includes a brief summary of this study, some managerial implications, and suggestions for evaluating travel Web sites.

2. Study background

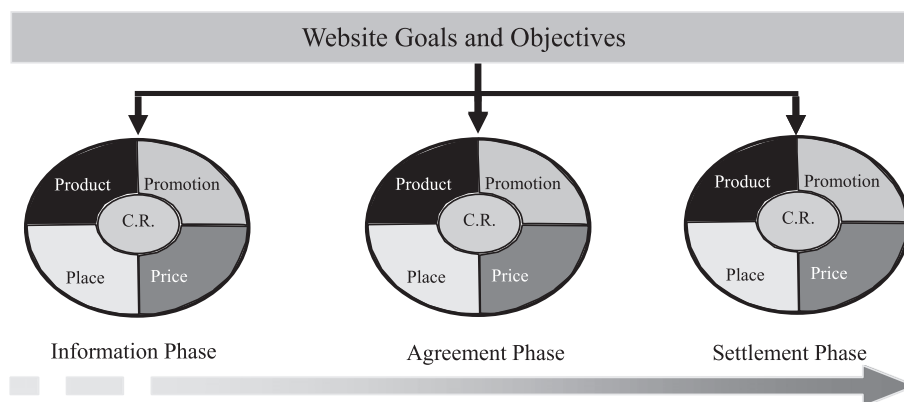
Various approaches in evaluating Web sites have been proposed by academic researchers since the late 1990s. The most common approaches include content analysis, benchmarking, survey, experiment, case study, and automatic evaluation. In particular, content analysis and benchmarking have been used widely by tourism and hospitality field researchers. For instance, Law et al. (2010) reviewed 75 papers, 27 of which adopted content analysis and 10 used benchmarking. These studies are generally divided into two broad categories: quantitative and qualitative. Many researchers are presently integrating quantitative and qualitative methods in their studies. These studies are further generally divided into five evaluation approaches: (i) counting, (ii) user judgment, (iii) automated, (iv) numerical computation, and (v) no actual evaluation. Law et al. (2010) also concluded that the adoption of a combination of methods provides a range of results that can satisfy the different needs of the entire range of stakeholders. Hence, what existing studies seem to have in common is a general agreement that assessing the effectiveness or performance of a Web site requires a multi-dimensional—rather than a unidimensional—approach or measure (Park & Gretzel, 2007).

Chiou, Perng, Tsai, and Lin (2008) selected 139 articles from 21 leading journals to identify trends in Web site evaluation and analyze frameworks and criteria proposed by different researchers. They classified these papers into three major categories: (i) information system (IS), (ii) marketing, and (iii) combination. In an IS-oriented study, 75% of the evaluation factors are technology related. On the other hand, in a marketing-oriented study, over 75% of the valuation factors, such as advertising, promotion, online transaction, order confirmation, and customer service, are marketing related. Combination-oriented studies feature a mixture of IS and marketing factors, and they have become prominent since the burst of the dot-com bubble.

After reviewing numerous evaluation frameworks introduced within the last decade, Chiou, Lin, and Perng (2010) identified three issues that should be further addressed. First, an evaluation framework must be process oriented to identify crucial activities in each transactional phase. For instance, Roney and Özturan (2006) proposed a process oriented framework for evaluating Turkish travel agencies. The framework consists of three levels of functionalities: corporate information, before-sale information, and sales and after-sales activities. Web site functionalities are analyzed through a business process perspective; hence, researchers and practitioners can effectively identify key customer-related activities. Second, a hybrid approach that considers the role of IS as a support factor in marketing, instead of a combined approach, suggests that IS factors should be embedded into marketing factors as facilitators of e-commerce. By doing so, the confusion in the classification of criteria can be eliminated. Third, existing studies have proposed various frameworks with extensive factors and criteria in evaluating Web sites. Unfortunately, none of these frameworks addresses the issue of the relationship between Web strategy and evaluation factors/criteria. A framework that considers strategy ensures that Web site presence is consistent with its pre-defined goals and objectives.

In response to these three issues, Chiou et al. (2010) proposed a strategic evaluation framework (Fig. 1). The framework is constructed based on the goals and objectives of each Web site. As mentioned earlier, the framework must be transactional process oriented. It should include information, agreement, and settlement phases. The information phase starts when prospective buyers enter the e-commerce system and lasts until they decide to place an order or leave the system. The agreement phase involves negotiations between prospective buyers and sellers, which are finalized by contracts. Eventually, the contracts are executed in the settlement phase according to stipulated conditions; product delivery and after-sales interactions take place during this phase. Chiou et al. (2010) collected representative criteria from 83 papers published in prestigious journals as the initial pool. To identify the most frequently used factors, the criteria were further classified into 12 unified factors as suggested by Park and Gretzel (2007). The top three factors are information quality, ease of use, and responsiveness.

After eliminating repetitive items, merging similar items, and condensing sub-attributes to higher-level criteria, Chiou et al. (2010) kept 53 criteria in the criteria pool (Table 1). Criteria in Table 1 bearing the superscript “T” are supported by information technology. This is an example of a hybrid concept. These criteria are categorized into five factors: product, promotion, price, place,



Source: Chiou, Lin, and Perng (2010)

Fig. 1. A strategic Web site evaluation framework.

Table 1
Criteria pool for Web site evaluation.

Factors/Criteria	Number of supported studies	Factors/Criteria	Number of supported studies
<i>Place</i>		<i>Product</i>	
Ease of navigation ^T	49	Product details	28
Content relevancy and usefulness	44	Product comparison ^T	13
Appealing and consistent style ^T	44	Product search or assortment ^T	10
Logical structure ^T	39	Product variety	10
Security protection ^T	38	Hierarchical product category	7
Ease of online transaction ^T	35	Product quality	4
User-friendly interface ^T	34	<i>Price</i>	
Comprehensive content coverage	33	Price details	14
Loading and processing speed ^T	32	Competitive price	5
Up-to-date content	31	All relevant charges details	5
Proper multimedia ^T	30	Price comparison ^T	4
Well and quick linkage ^T	29	<i>Promotion</i>	
Searching mechanism ^T	26	Promotion campaign	17
Easy to understand and read	27	Reputation and credibility of the site	15
Ease of access ^T	25	Company and brand recognition	13
Reliable and innovative system ^T	24	Purchasing guarantee	10
Accuracy	24	Advertising and banner	7
Easy to find target information ^T	22	<i>Customer Relationship</i>	
Online assistance and help ^T	16	Interactive communications ^T	37
Data retrieve mechanism ^T	14	Customized service ^T	28
Playfulness	13	Privacy policy	25
Convenient payment methods ^T	12	Quick response to customer ^T	25
Know the present location ^T	10	Customer service support ^T	23
Overview of selected items ^T	6	Member community ^T	19
Easy to cancel or modify order ^T	5	Order status inquiry and tracking ^T	17
		Valuable bundles or product suggestion	13
		Delivery product as promised	10
		Customized offerings ^T	9
		Convenient delivery options	8
		Ease of registration ^T	7
		Easy to return product	3

Source: Chiou et al. (2010)

^TRepresents the hybrid criterion supported by information technology.

and customer relationship (4PsC). These factors are used later for data analysis. The 4PsC factors are expressed in different shades to represent different relative impacts in three phases. The evaluation criteria are selected from a criteria pool in response to the goals and objectives of Web sites.

The strategic evaluation framework provides managers an internal evaluation mechanism to examine whether a Web site is consistent with its goal and objectives. The proposed framework is different from the frameworks of existing studies in two aspects. First, most existing frameworks are generally applicable to measurements of Web site usability, accessibility, design, quality, content, user satisfaction, user acceptance, and loyalty. The strategic framework adopts the goals and objectives of a Web site as guidelines in selecting relevant criteria to evaluate how well a Web site strategy has been accomplished. Second, most existing studies focus on the attitude and behavior of users toward the design and content of Web sites. The strategic framework uses the manager's viewpoint in examining the gap between "what the manager wants" and "what a Web site is" according to experts.

3. Methodology of the five-stage evaluation process

Evans and King (1999) suggested that any assessment tool has five components: categories (broad areas to be investigated), factors (specific elements comprising each category), weights (importance placed on factors), ratings (scores assigned to each factor), and weighted scores (an overall compilation based on both weights and ratings). Based on this concept, a hierarchical structure of Web site strategy is introduced to delineate the relationship between each category (strategic objectives) and relevant factors (criteria). In determining criteria weights, fuzzy linguistic terms are

selected to express the semantic decision-making process of evaluators. The rating of each criterion is determined by the group decision of a panel of experts. Lastly, a performance matrix chart is introduced to identify strategy-inconsistent criteria. The following section explains the steps involved in each stage of the evaluation process.

Stage one: Identification of Web site strategy and criteria.

Step 1: Identifying the goals and objectives of Websites. A personal in-depth interview with managers is suggested. A goal is a broad vision of a site that provides a general description of itself, such as "travel site with the best service quality." Objectives, on the other hand, are those that accomplish the goal of the Web site, such as "providing the most competitive prices."

Step 2: Selecting relevant criteria with regard to an objective. For instance, a travel Web site sets objective i (noted as O_i) as "providing customers with a variety of tour package selections," and the related activity is "aligning with other business organizations to introduce special tours, such as firework festivals or mountain biking." A related criterion j (noted as C_{ij}), such as "valuable bundles or product suggestions" (Table 1), is selected as a pertinent criterion. In an effort to confirm that a criterion is fully representative of an objective, group selection is performed to attain consensus between the researcher and the manager.

Step 3: Constructing a hierarchical evaluation structure. The structure is designed to delineate the relationship between the goals and objectives of Web sites and their related criteria. A hierarchical evaluation structure is helpful in cause-and-effect analyses.

Step 4: Assigning weights to each criterion. Managers thoroughly understand the goals and objectives of their Web sites; hence, they are asked to rate the importance of each criterion. Seven linguistic terms are used: “very unimportant,” “unimportant,” “somewhat unimportant,” “neutral,” “somewhat important,” “important,” and “very important”. Linguistic terms, instead of the Likert scale, are used because the assignment of criterion importance involves the uncertainty and fuzziness of human decision-making. Zadeh (1995) suggested that the fuzzy theory is more pertinent in delineating fuzzy characteristics while people are making judgments. In fact, fuzzy data may be expressed in linguistic terms or in fuzzy numbers. In transforming linguistic terms into crisp numbers, Chen and Hwang (1992) proposed a simple and effective methodology to solve fuzzy multiple attributes of the decision-making problem. According to the conversion scales of fuzzy numbers, these fuzzy linguistic terms can be transformed into one of the following crisp weights (W_{ij} , where i is an objective and j is its related criteria): 0.09, 0.23, 0.36, 0.50, 0.64, 0.78, and 0.91.

Stage two: Web-based evaluation instrument development.

Step 1: Transforming criteria into questions. The listed criteria are conceptual and general. Therefore, researchers must transform them into specific and practical questions that fit the Web site and the corresponding industry.

Step 2: Developing a Web-based questionnaire instrument. The online instrument includes two sections. Section I includes a summary of the interview with the Web site manager, and the hierarchical structure of the goal, objectives, and evaluation criteria of the Web site. This section aims to provide a background of the Web site and a reference for evaluators. Section II is designed for rating scores from the questionnaires. To improve the quality of the results and save time, an evaluation-supporting tool is provided at the end of each question. Human-based evaluation has been proven to be time consuming and inconsistent in testing certain features (Law et al., 2010). To address this problem, two types of evaluation-supporting tools are introduced: *hyperlink* and *finding*. A *hyperlink* is designed to direct evaluators to a relevant Web page. On the other hand, a *finding* is designed to present results on a specific question. For instance, a *hyperlink* is provided for the question “Does this Web site list functions in a logical order?” When an evaluator clicks on this *hyperlink*, a Web site home page opens so that the evaluator need not open a browser and look for the relevant Web page. An example of *finding* is the question “Does this Web site respond to customer questions effectively?” To answer this question, the authors pose the question “How is an order cancelled online?” on the customer service page of the Web site and record the response time. When an evaluator clicks on the *finding* icon, a results page shows the posted time, respondent time, and respondent answer to the question. Once online questionnaires and the evaluation-supporting tool are ready, a pilot test is necessary to identify potential problems of the instrument and provide a preview of how difficult the questions are to complete.

Stage three: Execution of Web site evaluation.

Step 1: Selecting a panel of experts as evaluators. As mentioned earlier, this paper aims to evaluate a Web site from the internal perspective of an organization to ensure that strategy is consistent with Web site presence. Law et al. (2010) also suggested that consumers do not have sufficient insights into how Web site performance is accurately measured. For these reasons, the expert-based evaluation

approach is used to assess the consistency of a Web site strategy. To make a proper group decision in rating the criteria, Robbins (1994) suggested that there should be five to seven experts.

Step 2: Rating each criterion. After identifying themselves on the Web-based instrument, evaluators rate each criterion using a linguistic term; they express their agreement or disagreement with statements under each criterion. The linguistic terms are “strongly disagree,” “disagree,” “somewhat disagree,” “neutral,” “somewhat agree,” “agree,” and “strongly agree”. As mentioned earlier, these fuzzy linguistic terms can be transformed into one of the following crisp scores (S_{ijk} , where i is an objective, j is its related criteria, and k is an evaluator): 0.09, 0.23, 0.36, 0.50, 0.64, 0.78, and 0.91.

Stage four: Criteria weights and score calculation.

Step 1: Normalizing criteria weights. Criteria weights are normalized to compare conveniently the relative importance of criteria under each objective. The normalized weight (NW_{ij}) is calculated as follows:

$$NW_{ij} = \frac{W_{ij}}{\sum_{j=1}^n W_{ij}} \quad (1)$$

where W_{ij} is the weight of criterion j with respect to an objective i , and n is the number of criteria j under an objective i .

Step 2: Calculating average scores, weighted scores, and objective scores. We need to aggregate individual judgments in a group into a single representative judgment. According to Saaty (1980), the geometric mean—not the frequently used arithmetic mean—accurately represents the consensus of experts, and it is the most widely used in practical applications. In this sense, the geometric mean is used to aggregate the judgments of a group of n evaluators. The average score of a criterion (AS_{ij}) is computed as follows:

$$AS_{ij} = \sqrt[n]{\prod_{k=1}^n S_{ijk}} \quad (2)$$

where n is the number of evaluators.

The weighted score of criterion j (WS_{ij}) and the weighted score of an objective (OWS_i) are calculated using the following equations:

$$WS_{ij} = AS_{ij} \times NW_{ij} \quad (3)$$

$$OWS_i = \sum_{j=1}^n WS_{ij} \quad (4)$$

where n is the number of criteria j under an objective i .

Stage five: Web strategy consistency analysis.

Step 1: Gap value analysis for each criterion. For purposes of improvement, managers should give attention to criteria with low average scores. However, in considering scores only as the basis for improvement, managers may be misled into allocating more resources to low-scoring criteria. To generate a more applicable assessment result, a gap (G_{ij}) is calculated by

$$G_{ij} = AS_{ij} - W_{ij} \quad (5)$$

where i is an objective and j is its related criteria.

The gap of each criterion is then compared with a threshold, which is decided by the manager, to determine whether such criterion is inconsistent with strategy. When the absolute value of

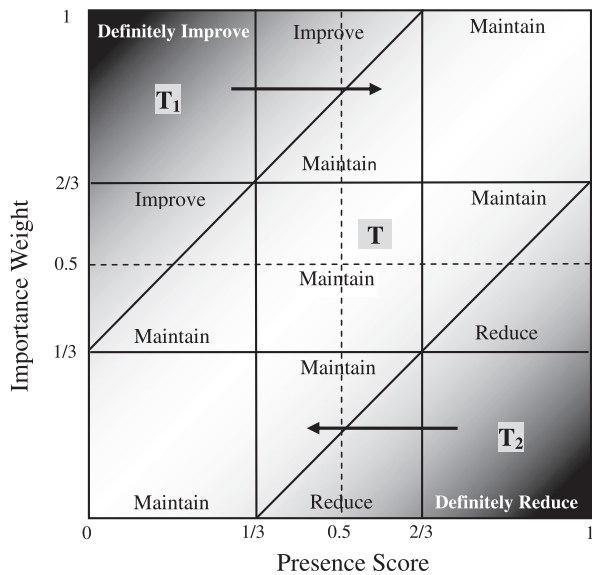
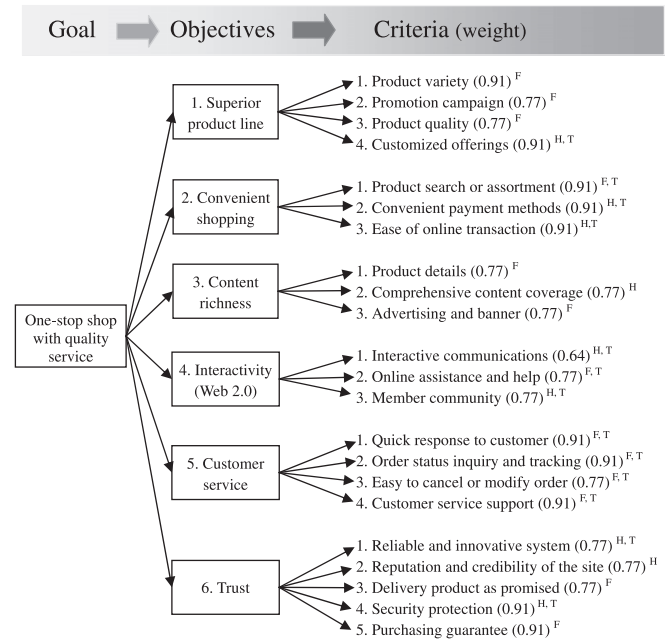


Fig. 2. Criteria performance matrix chart.

the gap is greater than the threshold, the criterion is recognized as inconsistent with strategy. However, a gap threshold is a subjective value and is adjustable based on the available resources and main concerns of a company.

Step 2: Constructing a criteria performance matrix chart. A matrix chart is used to provide managers with criteria performance information in graphical form. The matrix is also used in the priority ranking of improvement plans for strategy-inconsistent criteria. The criteria performance matrix (Fig. 2) originated from the service quality performance matrix (Hung, Huang, & Chen, 2003). A criterion presence score is plotted against the x-axis and the corresponding criterion importance (weight) is plotted against the y-axis. The scale of both axes is between 0 and 1. The matrix chart has nine cells when the axes are equally divided into three sections with four scales (i.e., 0, 1/3, 2/3, 1). Two off-diagonal lines, which are subjectively adjustable according to the organization's available resources, are added as a confidence interval to make the objective zone (T) more reasonable. The manager must determine a proper confidence level before interpreting the results. The objective zone (T) is defined as the area between two off-diagonal lines. A criterion located in the objective zone is consistent with strategy under a level of confidence and should therefore be maintained. A criterion located in the improve zone (T₁)—suggesting that the corresponding criterion presence score is far behind its importance weight—is considered inconsistent with strategy and should therefore be improved. To some extent, a criterion located in the reduce zone (T₂) is deemed consistent with strategy. However, managers should consider reducing the resource allocation for this criterion because its presence score far exceeds the importance weight (i.e., the criterion is over-performing). Criteria located at the top-left corner or the bottom-right corner (the darkest area) need improvement for better scores and reduction in resource allocation.

Step 3: Analyzing 4PsC dimensions and transactional process performance. The selected criteria are classified into the 4PsC dimensions following the criteria pool classification in Table 1. Each dimensional criteria weight (W_{dj}) and average score (AS_{dj})



^F search finding provided; ^H hyperlink provided; ^T technology supported

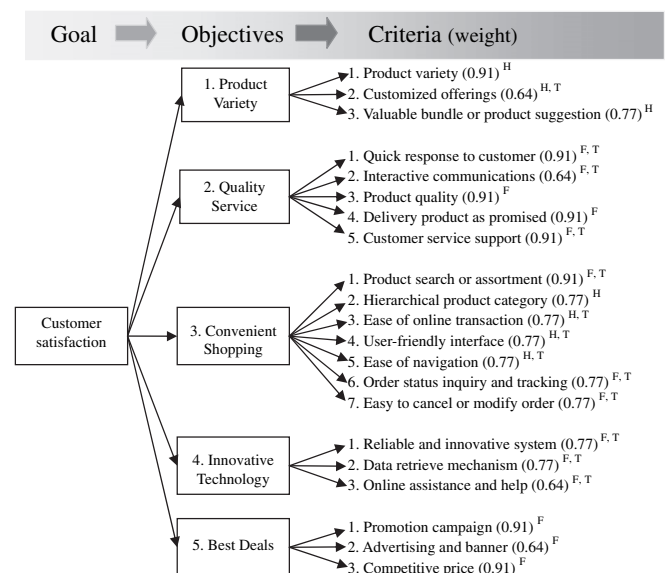
Fig. 3. Hierarchical evaluation structure and criteria weights of L site.

are added to compute the average dimensional weight (AW_d) and dimensional score (AS_d) using the following equations:

$$AW_d = \frac{\sum_{j=1}^n W_{dj}}{n} \quad (6)$$

$$AS_d = \frac{\sum_{j=1}^n AS_{dj}}{n} \quad (7)$$

where d is a 4PsC dimension ($d = 1-5$), j is a criterion number, and n is the number of criteria under the 4PsC dimension.



^F search finding provided; ^H hyperlink provided; ^T technology supported

Fig. 4. Hierarchical evaluation structure and criteria weight of Z site.

A radar chart is then constructed to identify the strengths and weaknesses of the 4PsC dimensions. By comparing the weights and scores of each dimension, we can identify the worst dimension, which is very important, but with poor presence. To analyze further the cause of the poor presence of the dimension, a drill-down analysis can help managers discover the related criteria with a relatively low score. Web customers leave a Web site without completing a transaction for many reasons. The analysis of each transactional phase can help businesses investigate which weak phase results in the loss of Web customers. To examine criteria presence in each transactional phase, the criteria (j) are first classified into one of three transactional phases (t) based on discussions with the manager and the definition of each phase. Subsequently, the criteria are further grouped into the 4PsC dimensions (d) following the criteria pool classification in Table 1. In this way, the average 4PsC dimensional weight (AW_{td}) and average score (AS_{td}) in each phase can be calculated following Eqs. (8) and (9), respectively.

$$AW_{td} = \frac{\sum_{j=1}^n W_{tdj}}{n} \quad (8)$$

$$AS_{td} = \frac{\sum_{j=1}^n AS_{tdj}}{n} \quad (9)$$

where t is the transactional phase ($t = 1-3$), d is a 4PsC dimension ($d = 1-5$),

j is the criterion number ($j = 1 \sim n$), n is the total criterion number under the 4PsC dimension in each phase, W_{tdj} is the weight of criterion j under a dimension d in phase t , and AS_{tdj} is the average score of criterion j under a dimension d in phase t .

4. Strategic evaluation of two Taiwan online travel sites

To demonstrate how the proposed evaluation framework can be implemented, we selected two leading travel companies in Taiwan

type of payment付款方式說明及下載

下載信用卡授權書
匯款、ATM轉帳
現金

\$ 下載信用卡授權書

- VISA CARD 授權書
- AE 授權書
- MASTER CARD 授權書
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Fig. 5. Hyperlink to payment method Web page of L site.

as study cases. *L* site has been in the traditional travel business for over thirty years and has been providing e-services to consumers since 2000. The company is both a travel agent and a supplier of travel products, such as flight tickets, accommodations, and group tour packages, to other agencies. The company's Web site is designed for the implementation of business-to-business (B2B) and business-to-consumer (B2C) Web strategies.

Another selected company is *Z* site. It has been in the travel business for nine years and was originally an online travel store. The company generally targets young clientele and families who intend to travel without tour guides. According to a survey by *InsightXplorer Limited*(2006), *Z* site is the most well known and frequently visited travel site in Taiwan. It also had the highest revenue and growth rate in Taiwan's online travel sector in 2006. These two companies have different business strategies for target customers and business operations; therefore, the evaluation results should provide good implications and comparisons.

4.1. Stage one: identification of web site strategy and criteria

After conducting an in-depth interview with *L* site's vice-general manager on October 14, 2008, the goal of *L* site was defined as "one-stop shop with quality service". To achieve this goal, six objectives are set: superior product line, convenient shopping, content richness, interactivity (Web 2.0), customer service, and trust. A hierarchical evaluation structure was constructed (Fig. 3) and sent back to the manager for confirmation. Fig. 4 shows the relationship between the goal, objectives, and criteria of *L* site. While the

Table 2
Weights, weighted scores, and gaps of *L* site.

Objectives (O_i)	Related Criteria (C_{ij})	W_{ij}	AS_{ij}	G_{ij}	NW_{ij}	WS_{ij}	OWS_i
1. Superior product line	1. Product variety	0.91	0.75	-0.16	0.27	0.20	0.66
	2. Promotion campaign	0.77	0.67	-0.11	0.23	0.15	
	3. Product quality	0.77	0.65	-0.13	0.23	0.15	
	4. Customized offerings	0.91	0.60	-0.31	0.27	0.16	
2. Convenient Shopping	1. Product search or assortment	0.91	0.74	-0.17	0.33	0.25	0.71
	2. Convenient payment methods	0.91	0.68	-0.23	0.33	0.23	
	3. Ease of online transaction	0.91	0.70	-0.21	0.33	0.23	
3. Content richness	1. Product details	0.77	0.66	-0.11	0.33	0.22	0.68
	2. Comprehensive content coverage	0.77	0.72	-0.05	0.33	0.24	
	3. Advertising and banner	0.77	0.66	-0.11	0.33	0.22	
4. Interactivity	1. Interactive communications	0.64	0.70	0.06	0.29	0.20	0.68
	2. Online assistance and help	0.77	0.64	-0.13	0.35	0.23	
	3. Member community	0.77	0.72	-0.05	0.35	0.25	
5. Customer service	1. Quick response to customer	0.91	0.70	-0.21	0.26	0.18	0.70
	2. Order status inquiry and tracking	0.91	0.71	-0.19	0.26	0.19	
	3. Easy to cancel or modify order	0.77	0.68	-0.09	0.22	0.15	
	4. Customer service support	0.91	0.70	-0.21	0.26	0.18	
6. Trust	1. Reliable and innovative system	0.77	0.52	-0.26	0.19	0.10	0.66
	2. Reputation and credibility	0.77	0.68	-0.09	0.19	0.13	
	3. Delivery product as promised	0.77	0.63	-0.14	0.19	0.12	
	4. Security protection	0.91	0.74	-0.17	0.22	0.16	
	5. Purchasing guarantee	0.91	0.69	-0.22	0.22	0.15	

Table 3
Weights, weighted scores, and gaps of *Z* site.

Objectives (O_i)	Criteria (C_{ij})	W_{ij}	AS_{ij}	G_{ij}	NW_{ij}	WS_{ij}	OWS_i
1. Product Variety	1. Product variety	0.91	0.70	-0.21	0.39	0.27	0.68
	2. Customized offerings	0.64	0.64	0.00	0.28	0.18	
	3. Valuable bundle or suggestion	0.77	0.70	-0.07	0.33	0.23	
2. Quality Service	1. Quick response to customer	0.91	0.66	-0.25	0.21	0.14	0.67
	2. Interactive communications	0.64	0.70	0.06	0.16	0.11	
	3. Product quality	0.91	0.64	-0.27	0.21	0.14	
	4. Delivery product as promised	0.91	0.64	-0.27	0.21	0.14	
	5. Customer service support	0.91	0.64	-0.27	0.21	0.14	
3. Convenient Shopping	1. Product search or assortment	0.91	0.72	-0.19	0.16	0.12	0.69
	2. Hierarchical product category	0.77	0.71	-0.06	0.14	0.10	
	3. Ease of online transaction	0.77	0.70	-0.08	0.14	0.10	
	4. User-friendly interface	0.77	0.70	-0.08	0.14	0.10	
	5. Ease of navigation	0.77	0.63	-0.14	0.14	0.09	
	6. Order status inquiry and tracking	0.77	0.68	-0.09	0.14	0.09	
	7. Easy to cancel or modify order	0.77	0.66	-0.11	0.14	0.09	
4. Innovative Technology	1. Reliable and innovative system	0.77	0.70	-0.07	0.35	0.25	0.69
	2. Data retrieve mechanism	0.77	0.68	-0.09	0.35	0.24	
	3. Online assistance and help	0.64	0.68	0.04	0.30	0.20	
5. Best Deals	1. Promotion campaign	0.91	0.72	-0.19	0.37	0.27	0.70
	2. Advertising and banner	0.64	0.70	0.07	0.26	0.18	
	3. Competitive price	0.91	0.68	-0.23	0.37	0.25	

hierarchical evaluation structure was being confirmed, the vice-general manager of *L* site was asked to assess the importance of each criterion using linguistic terms; the assessments were then transformed into crisp numbers. For instance, C_{11} (product variety)

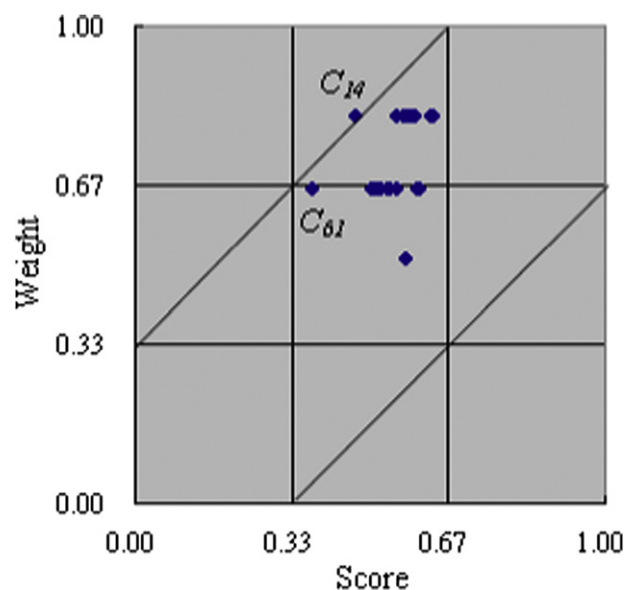


Fig. 6. Criteria performance matrix of *L* site.

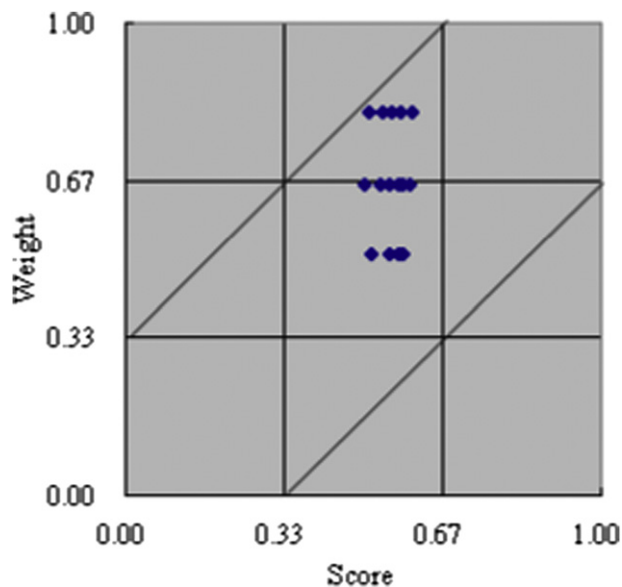


Fig. 7. Criteria performance matrix of Z site.

was assessed as “very important,” and the crisp number 0.91 was then assigned to W_{11} as suggested by Chen and Hwang (1992). Another interview with Z site’s marketing and IS manager was conducted on December 6, 2008. Accordingly, the goal of Z site is “customer satisfaction”. The site’s hierarchical evaluation structure is shown in Fig. 4. The criteria weights (W_{ij}) for L and Z sites are listed at the end of each criterion (Figs. 3 and 4, respectively).

The two sites substantially differ in several objectives. L site does not consider providing a “competitive price” to customers as a strategic criterion mainly because the company sells products not only to customers but to other travel agents as well. Hence, they need to reserve some margin for agents. In contrast, Z site is a pure online retailing shop, and one of its objectives is to provide the “best deals” to customers. Hence, C_{53} (competitive price) is included in the criteria list.

4.2. Stage two: Web-based evaluation instrument development

Identified conceptual criteria for the two Web sites were interpreted in a way that is relevant to them and to the tourism industry as a whole. Consequently, 22 questions were developed for L site, whereas 21 questions were developed for Z site. An evaluation-supporting tool was designed and provided at the end of each questionnaire as either a *hyperlink* or a *finding*. For instance, the question “Does the Web site provide a convenient payment method?” for L site has a *hyperlink* to the Web page of the payment method. The captured screen of L site’s Web page is shown in Fig. 5.

4.3. Stage three: execution of Web site evaluation

A panel of seven experts evaluated the criteria presence ratings. Three of the evaluators are e-business scholars in the tourism

Table 4
4PsC dimensional average weights and scores of L site.

Dimensions (d)	Related Criteria (C_{ij})	AW_d	AS_d
1. Place	$C_{21}, C_{22}, C_{23}, C_{32}, C_{42}, C_{61}, C_{62}, C_{64}$	0.84	0.68
2. Product	C_{11}, C_{13}, C_{31}	0.82	0.69
3. Price	N.A.	N.A.	N.A.
4. Promotion	$C_{12}, C_{33}, C_{53}, C_{65}$	0.67	0.81
5. Customer Relationship	$C_{14}, C_{41}, C_{43}, C_{51}, C_{52}, C_{54}, C_{63}$	0.83	0.68

Table 5
4PsC dimensional average weights and scores of Z site.

Dimensions (d)	Related Criteria (C_{ij})	AW_d	AS_d
1. Place	$C_{31}, C_{32}, C_{33}, C_{34}, C_{35}, C_{37}, C_{41}, C_{42}, C_{43}$	0.77	0.68
2. Product	C_{11}, C_{23}	0.91	0.67
3. Price	C_{53}	0.91	0.68
4. Promotion	C_{51}, C_{52}	0.77	0.71
5. Customer Relationship	$C_{12}, C_{13}, C_{21}, C_{22}, C_{24}, C_{25}, C_{36}$	0.79	0.67

industry, whereas the other four are owners and directors of local travel agencies. All the evaluators are well experienced in online travel transactions. Before evaluating each questionnaire, the evaluators identified themselves and read the goals and objectives of the two Web sites. While evaluating the questionnaires, evaluators could click either the *hyperlink* or the *finding* button at the end of each questionnaire, leading them to relevant Web pages or findings. Subsequently, evaluators used linguistic terms to express their agreement or disagreement with questionnaire statements.

4.4. Stage four: criteria weights and score calculation

We take the C_{11} (product variety) criterion of L site in Table 2 as a calculation example. The criterion weight (W_{11}) assigned by the manager is 0.91. The normalized weight ($NW_{11} = 0.27$) was derived using Eq. (1); we divided the criterion weight (0.91) by the total weight (3.36) of the four criteria under O_1 (superior product line). The average score 0.75 (AS_{11}) of C_{11} was calculated using Eq. (2). Using Eq. (3), we obtained the weighted score ($WS_{11} = 0.20$). The objective weighted score was then derived ($OWS_1 = 0.66$) using Eq. (4). The details of the criteria weights and scores for L and Z Web sites are listed in Tables 2 and 3, respectively.

4.5. Stage five: Web strategy consistency analysis

The gaps (G_{ij}) listed in Tables 2 and 3 were calculated using Eq. (5). Results showed that most of the criteria of the two Web sites have negative values. In theory, a negative gap shows that the criterion importance weight exceeds the corresponding presence score; hence, the criterion is deemed inconsistent with strategy. To determine the gap thresholds, Tables 2 and 3 were reviewed by the managers of L and Z sites, and the values were set as 0.23 and 0.25, respectively. When a criterion’s absolute gap is greater than the threshold, the criterion is considered inconsistent with strategy.

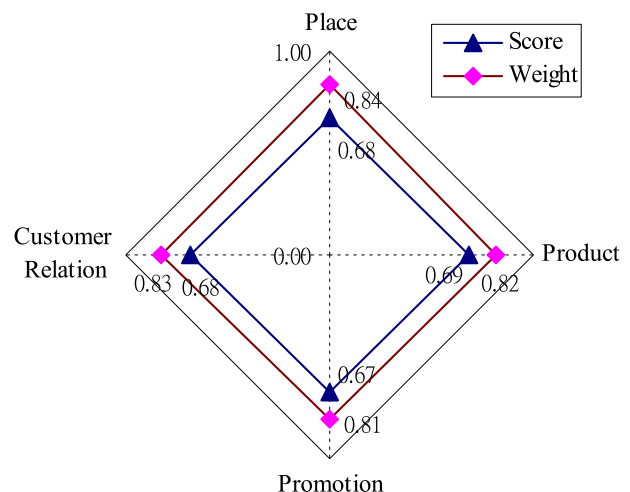


Fig. 8. 4PsC dimensions performance of L site.

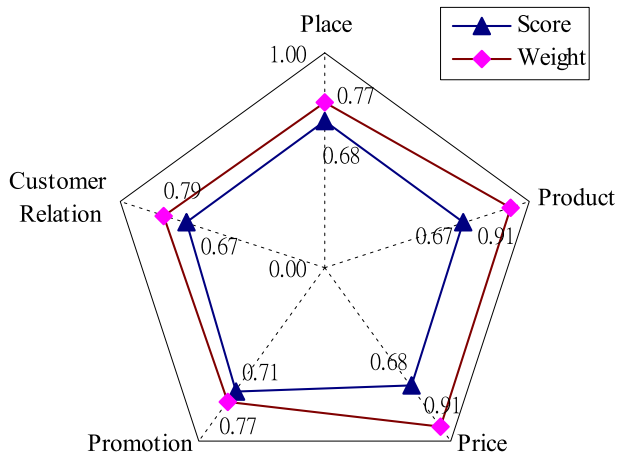


Fig. 9. 4PsC dimensions performance of Z site.

Based on the gap threshold, the three worse criteria of *L* site are C_{14} (customized offerings; -0.31), C_{61} (reliable and innovative system; -0.26), and C_{22} (convenient payment methods; -0.23). These criteria should be prioritized in action plans for improvement. For *Z* site, four criteria reach the gap threshold: C_{23} (product quality; -0.27), C_{24} (deliver product as promised; -0.27), C_{25} (customer service support; -0.27), and C_{21} (quick response to customers; -0.25). Although the two Web sites have a few criteria with positive gap values, no reduction in resource allocations should be taken because, as indicated by the managers, these gaps are reasonable.

To provide managers with a graphical analysis, a criteria performance matrix was introduced to identify criteria located outside the objective zone. Compared with the gap value method, the criteria performance matrix provides a better overview of criteria performance and can easily identify critical criteria that must be carefully watched. In Fig. 6, C_{14} (customized offerings) is located outside the objective zone and must be improved. The graphical analysis is consistent with the previous gap value analysis showing that the criterion has the largest negative gap value (-0.31) and is considered as the first priority for improvement.

Although C_{61} (reliable and innovative system) is still inside the objective zone, it is nearly located in the improve zone and should be closely watched. The same implication applies to *Z* site (Fig. 7). There are only three criteria weights that appear in both matrices because the weights were assigned by only one manager for each Web site. Both managers considered the selected criteria as “partly important”, “important”, and “very important”. Thus, only three crisp values were transformed accordingly. Broadly speaking, most criteria of the two Web sites are located in the objective zone, which means that the sites are generally consistent with their strategies. *Z* site seems to have a better performance than *L* site because its criteria scores are located near the center of the objective zone.

To identify the weaknesses and strengths of each 4PsC dimension by radar chart, the average dimensional weights (AW_d) and average dimensional scores (AS_d) were calculated using Eqs. (6) and (7), respectively. The results are summarized in Tables 4 and 5. As shown in Fig. 8, *L* site does not have all 4PsC dimensions; the “price” dimension is left out. This is because the site does not emphasize a pricing strategy and no pricing criterion is selected. The figure shows that the presence scores of the four dimensions are generally less than the weights by about 0.1 point. From a “product” standpoint, for example, the company may need to make additional efforts on low-scoring criteria, such as C_{13} (product quality) and C_{31} (product detail).

On the other hand, the evaluation of *Z* site entails a five-dimension analysis (Fig. 9). The manager regards “product” and “price” as very important dimensions. Both have the same weight of 0.91, but the scores of these two dimensions are moderate (0.67 and 0.68, respectively). When compared, the gaps between the scores and weights of these two dimensions are -0.23 and -0.24 , respectively, which are much higher than the gaps of the other three dimensions (-0.09 , -0.12 , and -0.06). Therefore, the manager should exert extra efforts in improving low-scoring criteria under the “product” and “price” dimensions, such as C_{23} (product quality) and C_{53} (competitive price).

To analyze the presence scores of 4PsC dimensions in three different transactional phases, we classified the dimensional criteria into these phases. The average 4PsC dimensional weights (AW_{td}) and scores (AS_{td}) in each phase were calculated using Eqs. (8)

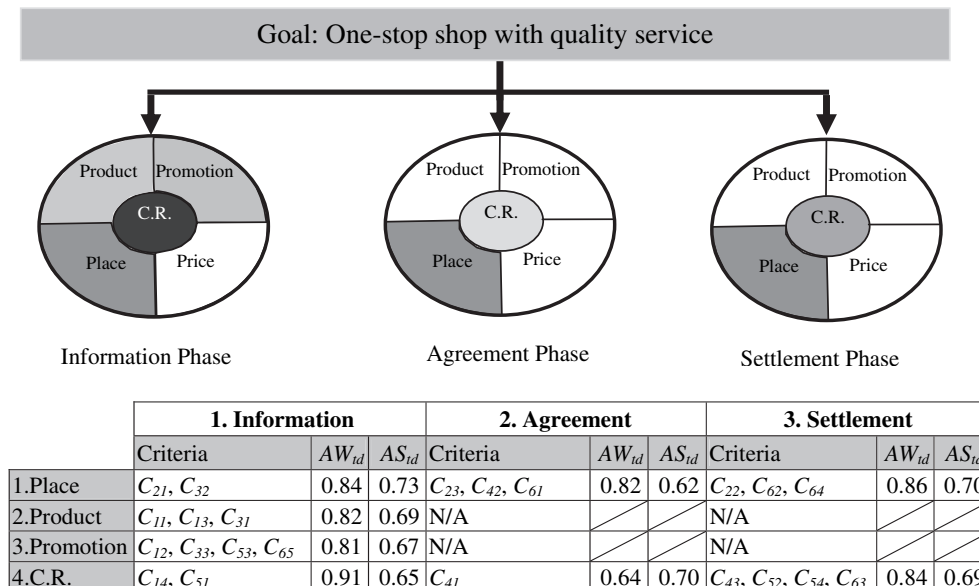


Fig. 10. *L* site's 4PsC dimensional average weights and scores in three phases.

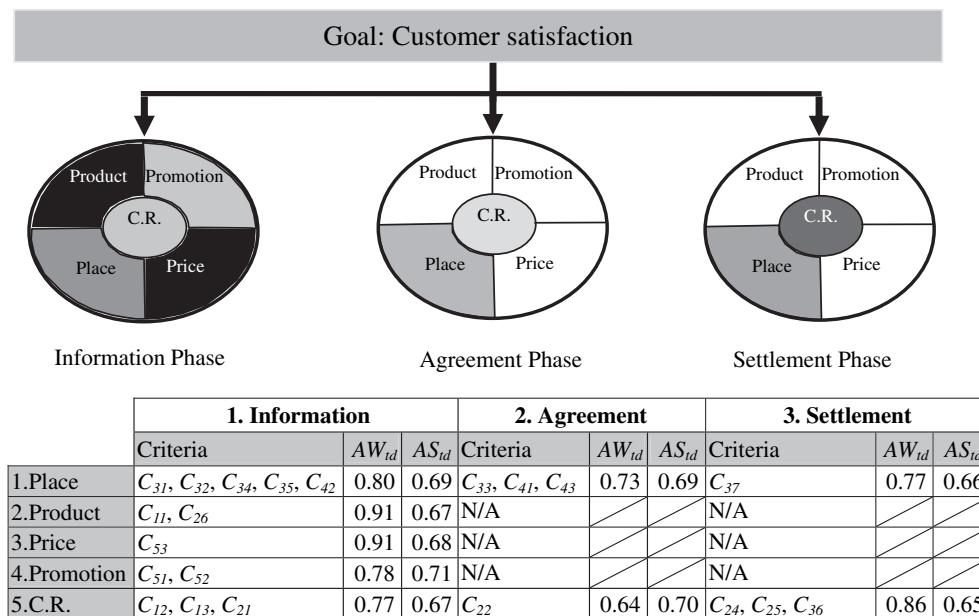


Fig. 11. Z site's 4PsC dimensional average weights and scores in three phases.

and (9), respectively. As shown in Fig. 10, “place” is the best-performing dimension of *L* site with the highest average score of 0.73 during the information phase, whereas “C. R.” is the worst-performing dimension with a low average score of 0.65. Criteria under the “place” dimension mostly relate to Web site design and information technology. A user-friendly Web site can help customers find target information easily and decide quickly. In exploring the reason for the low “C. R.” score, we found that C_{14} (customized offerings) was rated with an average score of 0.6 (Table 2, AS_{14}) by the seven evaluators. This criterion can be improved by offering customers more differentiated travel products and flexibility, enabling them to customize their own tour plans. Note that pricing is not considered by *L* site; hence, “price” is not presented in the information phase. In the information phase, the manager considers “C. R.” as the most important dimension (the darkest shade) with a weight of 0.91, but the score of this dimension is only 0.65. In this phase, this dimension is regarded as inconsistent with strategy. From the viewpoint of the agreement phase, the “place” dimension has the lowest average score (0.62). To address further this issue, two problem criteria were identified: C_{42} (online assistance and help) and C_{61} (reliable and innovative system). These criteria have low scores of 0.64 (Table 2, AS_{42}) and 0.52 (Table 2, AS_{61}), respectively.

Fig. 11 shows the dimensional scores of *Z* site in the three transactional phases. Compared with *L* site, *Z* site includes “price” as one of the five-dimensions in the information phase; C_{53} (competitive price) is considered a very important criterion in the site's pricing strategy. “Promotion” is the best-performing dimension with a high score of 0.71, which is different from *L* site's best-performing dimension, that is, “place,” in the information phase. This can be explained by the fact that *Z* site is a pure online travel company and competitive pricing is a key factor in attracting online customers. Periodically introducing attractive promotion campaigns is important for *Z* site to create more sales because the site does not sell products to other travel agencies, unlike *L* site. Indeed, the manager considers “product” and “price” as very important dimensions (the darkest area) in this phase, but the scores of these two dimensions are low. It appears that in these two dimensions, an inconsistency exists between Web site strategy and

presence. Lastly, “C. R.” is the worst dimension, with a low score of 0.65 in the settlement phase—because the site performs poorly in delivering products as promised and in providing good customer service support.

5. Conclusions and suggestions

Very limited research explores the issue of Web strategy in Web site evaluation, and none includes Web strategy in their evaluation frameworks. In response to this, a strategic framework was adopted to ensure consistency between Web strategy and actual Web site presence. To select relevant criteria from the proposed criteria pool, a strategic framework was developed based on the goals and objectives of Web sites. To implement this framework, we introduced a five-stage evaluation process, serving as a systematic approach to strategic Web site evaluation.

To illustrate how a Web site evaluation is conducted using a strategic framework, this study selected two leading travel Web sites in Taiwan as demonstration cases. Two most telling findings emerge from the study. First, *L* and *Z* sites are apparently different in their Web site strategies; hence, criterion selection not only varies in the number of criteria but also in criterion attributes. Figs. 3 and 4 clearly demonstrate differences in the evaluation frameworks of these sites. Second, our Web strategy consistency analysis assists managers in identifying inconsistent areas and in improving such areas. For instance, *L* site has the goal “one-stop shop with quality service,” and the objective “convenient shopping” is regarded as one of its relevant objectives. We examined whether this objective is consistent with the goal. Table 2 indicates that this objective is in line with the goal because it obtained the highest score (0.71). Nevertheless, the manager can still introduce improvements to this objective by providing more convenient payment methods. *Z* site, on the other hand, argues that “quality service” is a necessary objective in attaining its goal, which is “customer satisfaction.” Unfortunately, this objective obtained the lowest score (0.67). To remedy this inconsistency issue, the manager can introduce improvements to relevant low-scoring criteria in Table 3, such as C_{21} (quick response to

customer), C_{23} (product quality), C_{24} (deliver product as promised), and C_{25} (customer service support).

We introduced an evaluation-supporting tool to provide a context reference for evaluators through a hyperlink of relevant Web page or supporting materials with respect to specific questions. By doing so, evaluators can easily and quickly focus on target Web pages or refer to facts. Nevertheless, evaluators must bear in mind that this supporting tool acts as a decision-making reference. To make objective evaluations, evaluators are free to explore more Web pages or online transaction processes when necessary. To improve the quality of the supporting tool, researchers or practitioners may collect suggestions from evaluators to redesign the hyperlink or finding on relevant questions.

This research has a twofold contribution. First, the strategic framework helps managers easily develop individual Web site evaluation criteria based on their strategies. The strategic framework was designed to focus on how a specific Web site applies its goals and objectives, not for an industry or the Internet in general, as designed in previous studies. Second, the five-stage evaluation process assists manager in effectively discovering strategy-inconsistent criteria through a combination of qualitative and quantitative methods.

There were two limitations of this study. First, the 4PsC dimensions apply to commercial Web sites but not to non-commercial Web sites. The “price” dimension may not be relevant to non-commercial Web sites; it can therefore be removed accordingly. Second, the evaluation criteria pool may need to be revised by collecting relevant criteria with respect to non-commercial Web sites. Furthermore, rapid developments in information technology have introduced a plethora of new terminologies. The criteria listed in Table 1 are conceptual and can be further interpreted to fit specific industries and individual contexts.

As suggested by Law et al. (2010), seeking the views of industrial practitioners and consumers remains important because these groups are the ultimate suppliers and users of tourism Web sites. We agree that an external evaluation is still necessary in assessing the effectiveness of Web sites from the consumer viewpoint. This should be conducted after the strategy consistency of Web sites is confirmed through our proposed internal evaluation process. If the two evaluation results show significant differences in particular criteria, managers may need to review their strategies and clarify the causes of such discrepancies.

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